

## ENGLISH SENTENCE STRESS, GRAMMATICAL FUNCTIONS AND CONTEXTUAL COREFERENCE

### 1.0 Introduction\*

The goal of this paper is to provide a better understanding of the dynamics of English sentence stress by examining it within a discourse grammar framework. It will be maintained that the position of sentence stress is governed by two interacting factors: a hierarchy of grammatical functions and the coreference status of the lexical items realizing these functions with respect to previous parts of the discourse. This approach differs from previous generative treatments, where sentence stress is described solely with reference to the syntactic constituent structure (Chomsky & Halle 1968; Bresnan 1971, 1972). Conceptually, the present model is influenced by the Praguean Functionalist Sentence Perspective, an approach which has not been without impact on American linguists (Chafe 1974:119–29; Liberman 1978:167).

We will first review some previous attempts that have been made to deal with English sentence stress in the phonological literature before presenting our own approach. Most analyses within the generative framework have been based on the assumption that it is possible to predict the relative degrees of suprasegmental prominence within a sentence using only information from the labelled bracketed surface structure.

### 1.1 Chomsky & Halle (1968)

In Chomsky & Halle (1968), for example, it is maintained that sentence stress is assigned by the Compound Rule and the Nuclear Stress Rule, which can be formalized as in (1) (p. 18):

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$$(1) \quad \left[ \begin{array}{c} 1 \text{ stress} \\ V \end{array} \right] \rightarrow [1 \text{ stress}] / \left\{ \begin{array}{c} \text{---} \cdots \overset{1}{V} \cdots \\ 1 \\ V \cdots \text{---} \cdots \end{array} \right\}_{\text{NAV}} \quad \begin{array}{l} \text{a) Compound Rule} \\ \text{b) Nuclear Stress Rule} \end{array}$$

The effect of rule (1) is to place primary stress on the first stressed vowel of a compound (e.g. *blackboard*) and on the last stressed vowel of a phrase (e.g. *black board*), each word having previously been assigned stress by rules of word phonology. Furthermore, by a general convention of stress assignment, each time an element receives primary stress, all other stresses within the domain of application are reduced by one. Rule (1), along with all other stress rules, have been assumed to apply cyclically to surface structures. As an example of how the two ordered subrules of (1) interact to produce a prominence contour, the sentence *John's blackboard eraser was stolen* is analyzed in (2) (see Chomsky & Halle 1968:22):

$$(2) \quad \begin{array}{ccccccc} s_{[NP[D[John's_D]N[N[A[black_A]N[board_N]N[N[eraser_N]N]NP]VP[V[[was][stolen_V]VP]s]]} \\ \begin{array}{ccccccc} 1 & & 1 & & 1 & & 1 \\ & & 1 & & 2 & & \\ & & 1 & & 3 & & 2 \\ 2 & & 1 & & 4 & & 3 \\ 3 & & 2 & & 5 & & 4 \end{array} & & 1 & & & & 1 \end{array} \quad \begin{array}{l} (1a) \\ (1a) \\ (1b) \\ (1b) \end{array}$$

One problem associated with the SPE system is that, due to the convention of stress reduction and to the fact that the NSR can apply to its own output, complex sentences will receive prominence contours containing indefinitely weak stress, such as the sentence mentioned by Chomsky & Halle (1968:23): *My friend can't help being shocked at anyone who would fail to consider his sad plight*, where *sad plight* would end up with the contour *sad plight* after 7 applications of the NSR. Chomsky and Halle (1968:23) note this shortcoming of their analysis and 'make it quite clear that the rules discussed above give accurate results only for very simple constructions'. They suggest, furthermore, that 'the problem of extending this description to a wider class of cases may be nontrivial' (1968:24).

The problem involving multiple degrees of stress was avoided in later phonological descriptions by working on a more abstract level of metrical structure with representations consisting of structures with alternating patterns of S[trong] and W[eak] syllables. The problem of determining sentence stress still remains, however (see below, § 3.3.), since metrical trees are, like Chomsky and Halle's stress contours, based on the syntactic surface structure and the principles embodied in the Compound and Nuclear Stress Rules.<sup>1</sup>

## 1.2 Bresnan (1971)

In what was the first detailed generative treatment of English sentence stress, Bresnan (1971) pointed out that there was a large class of exceptions to the NSR. For example, under normal conditions, the following sentence-final items do not receive final stress (258):

(3) Anaphoric pronouns:

- (a) Robert discussed it<sup>1</sup>
- (b) \*Robert discussed it<sup>1</sup>

(4) Indefinite pronouns:

- (a) The girl picked some<sup>1</sup>
- (b) \*The girl picked some<sup>1</sup>

Other anaphoric elements, even when grammatically definite, are not assigned sentence stress:

- (5) Mary knows a man<sub>i</sub><sup>1</sup> who excels at judo, and she adores the man<sub>i</sub>.

As regards these exceptions to the NSR, Bresnan assumes that 'by some means or another, anaphoric and indefinite elements are not assigned primary stress' (1971:258). She does not deal with the derivation of the stress pattern of such sentences, but instead attempts to account for another set of apparent exceptions to the NSR. These involve sentences which can be analyzed as having undergone movement or deletion of NP objects, e.g.:

- (6) (a) George has plans to leave<sup>1</sup>  
George has plans [<sub>S</sub>George leave plans<sub>i</sub>]
- (b) George has plans to leave<sup>1</sup>  
George has plans [<sub>S</sub>George leave]
- (7) (a) Mary liked the proposal that George left<sup>1</sup>  
Mary liked the proposal [<sub>S</sub>that George left the proposal]

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<sup>1</sup> '...we simply define a relation on each pair of sister nodes in the syntactic structure, the output depending on certain local properties of that structure. To be specific, this theory's version of the NSR and CSR can be stated as follows:

(8) in a configuration [<sub>C</sub>AB]<sub>C</sub>:

- (a) NSR: If C is a phrasal category, B is strong
- (b) CSR: If C is a lexical category, B is strong iff it branches.' (Lieberman & Prince 1977:257.)



(10') \*Mary liked the proposal that was made<sup>1</sup>

Although they do not develop any alternative for describing English sentence stress, Berman and Szamosi propose the following surface generalization which they claim governs its placement to some extent; they note that 'regardless of the "derivational history" of a sentence, if it ends in an NP, this NP receives primary stress' (1972:309). They note, however, that there are a number of exceptions, most notably, anaphoric items, e.g.

(11) The children didn't want to go to bed, so<sup>1</sup>

(a) John scolded the governess<sup>1</sup>

(b) John scolded the bastards<sup>1</sup>

Moreover, Berman and Szamosi feel that the different stress contours of phrases like *plans to leave* (cf. (6)), corresponding to different meanings 'remain valid counterexamples to a theory of stress-assignment which relies solely on surface structure' (1972:310). Sentences like (12) are also exceptions to Berman and Szamosi's surface structure generalization, as was later pointed out by Bresnan (1972:331):

(12) Peter had clams for dinner<sup>1</sup>

Berman and Szamosi come to the conclusion, therefore, that 'neither the surface level application nor the cyclical application of the NSR works. Can the NSR be made to work at all? Probably not, especially because of the existence of a large number of cases which indicate that primary stress assignment (let alone the entire stress contours) often depends on factors that only marginally, if at all, involve structure' (1972:311–12).

## 1.4 Bresnan (1972)

In a reply to Berman and Szamosi, Bresnan (1972) admits that not only structural, but also semantic factors are involved in determining the placement of primary stress. She does not abandon the NSR, however, but rather proposes a new rule, that of Topical Stress Assignment, an optional rule which is assumed to apply before the NSR. This rule of Topical Stress Assignment is assumed to assign stress in the exceptional cases to the NSR. For example, Bresnan notes (1972:328) that in simple declarative sentences ending in a predicative, the primary stress is often most naturally placed on the subject:

- (13) (a) The sun is shining  
 (b) A proposal was made  
 (c) A new book has appeared  
 (d) An umbrella was lost  
 (e) Work must be done  
 (f) Guests arrived

Topical Stress Assignment is assumed to account for the patterning of sentence stress in (13). It is also assumed to handle Berman and Szamosi's counterexamples to the NSR, e.g.

- (10) Mary like the proposal that was made (A propósal was made)  
 (14) What books have arrived? (Some bóoks have arrived)  
 (15) There is work to be done (Wórk (must) be done)

The inclusion of a rule of topical stress assignment in addition to the NSR makes the grammar of sentence stress that much more complicated. It is all the more dissatisfying due to the fact that no details are presented as to the conditions of application of the Topical Stress Assignment Rule.

## 2.0 A proposal: Discourse Coreference and Grammatical Functions (DisCoGramFunc)

We think that Berman and Szamosi's idea that it is surface structure that is relevant to determining the place of sentence stress can be made to work if we take an 'enriched' surface structure as input. We are of the opinion that, to the extent that syntactic information is relevant to the assignment of sentence stress, it is grammatical functions, not constituent structure that provides the most insightful basis for understanding the dynamics of sentence stress. The position of sentence stress is, we would maintain, highly predictable on the basis of what we would propose is a natural hierarchy of grammatical functions based on their potential for attracting stress and the coreferential status of the lexical items realizing these functions with respect to preceding parts of the discourse.<sup>2</sup> In other words, we claim that the

<sup>2</sup> For the present, we will not attempt to formulate specific rules determining contextual coreference but rather follow Firbas' (1979:127) guide rule:

position of greatest prosodic prominence in a given English sentence within a discourse is based not on the NSR, but rather on a hierarchy of grammatical functions of the following type:

(16) Predicate Complement > Subject > Verb

That is to say, in a given sentence, the predicate complement<sup>3</sup> is most likely to receive stress if it is contextually noncoreferent; if it is contextually coreferent, then sentence stress will fall on the subject if it is contextually noncoreferent. If the subject is also contextually coreferent, sentence stress will fall on the verb.

This hierarchy is based on semantic notions. The fact that subjects and objects tend to receive more prominence than verbs has been observed by a number of scholars.<sup>4</sup> For example, Schmerling (1976:82) formulates the idea

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'An element is context dependent [i.e. contextually coreferent M.H.] if it conveys information derivable from the preceding verbal context and/or if it refers to a perfectly obvious item belonging to the situational context of immediate experience shared by the speaker and listener, context dependence or independence ultimately depending on the communicative purpose entertained by the speaker at the point that, at the moment of utterance, has been reached in the development of discourse. It should be evident that in regard to context dependence, context is here understood in the narrowest way.'

<sup>3</sup> By 'predicate complement' we mean, following Lyons (1969:345), a 'word or phrase (other than the verb itself) which is an obligatory constituent of the predicate: for instance, the object of a transitive verb ..., nominal or adjectival expressions which combine with the 'copula' in such sentences as *Mary is a beautiful girl* and *Mary is beautiful* ..., and such 'adverbial' expressions as *in Central Park* or *on Sunday* in sentences like *The parade was in Central Park* or *The demonstration was on Sunday*'. It will also be assumed that adverbials having selectional restrictions with the lexical verb are included under this category, for example, *go*: Locative adverbial, *behave*: Manner adverbial (see Allerton and Cruttenden 1978, Dik 1978, and Quirk *et al.* 1972). Further research is required as regards specification of these restrictions for individual verbs.

<sup>4</sup> It has been noted that, as regards child language, subjects and objects are acquired before verbs. It is interesting to speculate that this may be related to their greater likelihood of receiving prosodic prominence in discourse. McNeill (1970:66) notes, for example, that 'Greenfield observed the appearance of subjects and objects towards the end of the holophrastic period, but no verbs'. Moreover, in two-word sentences, the verb is often omitted, whereas the subject and/or object are always present. This is explained by McNeill (*ibid.*) as being due to the relational character of the verb:

'... all possible two-word combinations occur in child speech-subject-object, verb-object and subject-verb. However, these combinations occur with unequal frequency in Bloom's records, declining in the order given ... The most general interpretation of verbless sentences is to regard verbs as being words that stand for grammatical relations, and that, like prepositions, they tend to appear only after combinatorial patterns with non-relational words are established. Such an interpretation is consistent with the progress in child grammar, from the

in her Principle II: 'The verb receives lower stress than the subject and direct object, if there is one; in other words, predicates receive lower stress than their arguments, irrespective of their linear position in surface structure'. A similar observation is made by Firbas (1979:129):

'... in regard to a context independent object, subject complement, and adverbial functioning as a specification, not as a setting, the verb will as a rule not exceed any of these elements in CD [CD = *communicative dynamism*, a term within Praguean Functionalist Sentence Perspective which refers to the relative extent to which a given element contributes towards the amplification (development) of the discourse]. In the absence of these elements, it will not even exceed a context independent subject in CD. This is because in the development of discourse the primary function of the verb is an introductory one. It consists in introducing into the discourse notions conveyed by the context independent elements. This explains why the verb comparatively rarely comes to carry the highest degree of communicative dynamism and to function as the IC [IC = intonation center]'.<sup>1</sup>

The fact that the predicate complement precedes the subject in the hierarchy can be related to the fact that in English, the rheme (or new information) tends to be placed at the end of the sentence, the normal position of the predicate complement, whereas subject position tends to be reserved for the theme (or old information).

The hierarchy in (16) should, in fact, be further specified in order to account for the internal structure of a given phrase, since it is not always the case that it is automatically the *head* that receives the stress. In cases where it is only the head that is contextually coreferent and not the whole phrase, it is the *modifier* that will be assigned sentence stress, as in (17b):

- (17) (a) My mother gave me a [yellow [dress<sub>i</sub>]]<sub>j</sub>  
 (b) I really wanted a [blue [one<sub>i</sub>]]<sub>k</sub>

Compare this with (18b) where the whole object phrase is coreferential with *yellow dress*, thus causing sentence stress to be placed elsewhere:

- (18) (a) My mother gave me a [yellow [dress<sub>i</sub>]]<sub>j</sub>  
 (b) I'm going to wear [the gorgeous [thing<sub>i</sub>]]<sub>j</sub> today.

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holophrastic period on to express grammatical relations first through combinations and last through special words.'

McNeill does not take up prosodic prominence which certainly constitutes a factor making a particular grammatical function within a sentence more perceptually salient for the child and thus perhaps easier to learn.

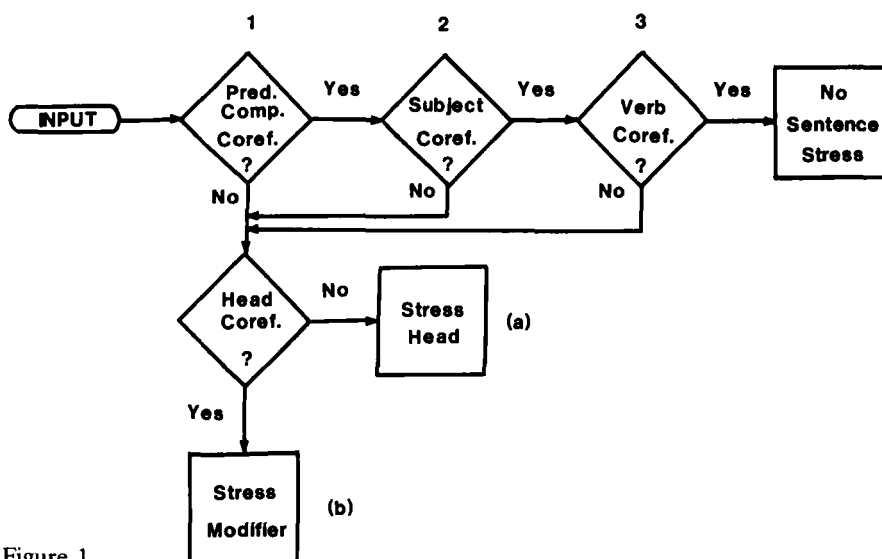


Figure 1

With these facts in mind, the process of assigning sentence stress can then be formally represented by means of a flowchart such as that in Figure 1.

Figure 1 is to be interpreted as follows:

To assign sentence stress in a given piece of dialogue, proceed in the following way: (1) check the Predicate Complement if there is one; if the whole phrase is noncoreferential with respect to the preceding context, assign the Head sentence stress if it is likewise noncoreferential (1a). If the head is coreferential, however, assign stress to the Modifier (1b). If the whole Predicate Complement is contextually coreferent on the other hand, go to (2), Subject. The same procedure is repeated here as in the case of the Predicate Complement. Should the procedure be carried to the Verb position and a 'Yes' response be obtained even here as regards the coreferential status of the whole verb phrase, then the sentence is marked 'No Sentence Stress'. Such a sentence is in some respect 'devient' as regards 'normal' sentence stress dynamics, e.g. a sentence which provides no 'new' information, such as (19b):

- (19) (a) Kids<sub>i</sub> like<sub>j</sub> icecream<sub>k</sub>  
 (b) They<sub>i</sub> like<sub>j</sub> it<sub>k</sub>

## 2.1 Comparison with NSR

It is interesting to note that the characteristic 231 stress pattern derived by the NSR for simple SVO sentences reflects the ordering in (16), e.g.:

(20)  $\begin{matrix} 2 & 3 & 1 \\ \text{Rats} & \text{like} & \text{cheese} \\ \text{s} & \text{v} & \text{o} \end{matrix}$

According to the NSR, however, all sentences receive sentence stress on the rightmost element containing [1 stress]. In reality, this is not the case as we have seen above, e.g.:<sup>5</sup>

(11) The children didn't want to go to bed, so

· (a) John scolded the governess

(b) John scolded the bastards

(13) (b) A proposal was made

Moreover, if one listens to a given piece of English dialogue, it becomes very difficult to accept Chomsky and Halle's (1968:25) claim that it is the listener's knowledge of the Nuclear Stress Rule that determines the stress contour of a given set of sentences within a context. One would tend to agree with Berman and Szamosi (1972:312) that it is perhaps misleading to carry over devices that worked well for the description of word-level phonology to the description of larger phrasal units and that 'some entirely novel ideas are necessary'.

## 2.2 Data

We have taped a number of episodes of the American television program 'Dynasty' in order to get some idea of how sentence stress patterns in English. Although it is too early to draw any definite conclusions, it would appear that the Nuclear Stress Rule fares very badly. A model such as that in (16') seems to work much better in the analysis of sentence stress. Following is an example of the patterning of sentence stress in a piece of discourse from 'Dynasty'. The word or words that bear most prosodic prominence in each sentence have been written in bold letters. The subscript *c* has been used to indicate coreferent items that are not immediately obvious to the reader, e.g. things that had been talked about before the present dialogue began. Instances of obvious anaphoric coreference are indicated by the subscripts *i* and *j*:

<sup>5</sup> For sentences cited in isolation, names appear to act as contextually coreferent, even if they are mentioned only once. This is a peculiarity of names that has been noted by several scholars (see Ladd (1978:90-1) for a discussion of this phenomenon).

- (21) Blake: (a) Dammit, Taft, we have a **crisis**<sub>i</sub> here.  
 (b) I **need** that information<sub>c</sub> within the next two hours.  
 (c) Now Taft, if the **sun** comes up on this one<sub>i</sub>  
 and it's<sub>i</sub> not **settled**  
 we can all meet on the **unemployment** line.  
 (d) What is the **latest** obituary<sub>c</sub> from our mortician, Taft?  
 Taft: (e) It seems **definite**.  
 (f) The revolutionary government has confiscated the **oil**.  
 (g) Your **reserves** have been nationalized.  
 Blake: (h) Are you **hurrying** with those<sub>c</sub> j?  
 Secretary: (i) As fast as I **can**, Mr. Laird.  
 Blake: (j) Andrew, I'm not **sure** I wanna go **through** with those  
 papers<sub>j</sub>  
 Taft: (k) **Why not**?  
 (l) It's a **gift**.  
 (m) There's no **law** that says a man can't give a **gift** to his  
**spouse**.  
 (n) Besides Blake, you have no other **choice**.  
 (o) I'm gonna have to **insist** that you **do** it.

The patterning of sentence stress in the dialogue in (21) seems to follow the hierarchy in (16). Thus, with reference to the flowchart in Figure 1, this patterning can be accounted for in the following way: in sentence (a), the object, a contextually noncoreferential noun, is stressed according to step (1a). In (b), however, the object, *that information*, has an antecedent earlier on in the dialogue and thus does not receive prominence. The subject, *I*, is also contextually coreferential. Consequently, the sentence stress falls on the verb, *need*, according to step (3a) in Figure 1. In the first conjunct in sentence (c), there is no predicate complement; consequently, the stress falls on the contextually noncoreferent subject, *sun*, according to step (2a). In the second conjunct, there is likewise no predicate complement; the subject, however, is coreferential with *crisis* in sentence (a); therefore, the stress falls on the verb, *settled* by (3a). In the main clause of sentence (c), the predicate complement, a noncoreferential locative adverbial, receives stress at (1a). In sentence (d), the predicate complement is not contextually coreferential; its head, however, *is* (i.e. it refers to 'news' discussed earlier in the dialogue); consequently, sentence stress falls on the modifier, *latest*, according to step (1b). The same patterning can be seen to exist throughout the remainder of the dialogue. We should point out here that we assume that each clause (S) of a sentence is submitted to analysis according to the model. Furthermore, we will assume that the assignment of sentence stress applies recursively at

each step in Figure 1 so as to account for instances where more than one stress is present (as in (21m)) (I owe this observation to Gösta Bruce).

We have also constructed possible texts where the sentence stress pattern follows the model in Figure 1. An example is found in (22), where identical subscripts indicate coreference. Numbers following the underlined words indicate at which point in Figure 1 sentence stress is applied. The text sounds very natural when read aloud:

- (22) My mother<sub>i</sub> sent me some **money**<sub>j</sub> (1a) yesterday.  
I really **needed** (3a) the cash<sub>j</sub>.  
I'd **spent** (3a) all my dough<sub>j</sub> and asked the dear<sub>i</sub> for a **loan**<sub>k</sub> (1a).  
My brother<sub>i</sub> was of course **livid** (1a).  
He<sub>i</sub> **refused** (3a) me a loan<sub>k</sub> earlier and the creep<sub>i</sub> called me a  
**spendthrift**<sub>m</sub> (1a).  
That's absolutely **ridiculous** (1a)!  
Now my **sister**<sub>n</sub>'s (2a) spendthrift<sub>m</sub>.  
She<sub>n</sub>'s a **real** (1b) squanderer<sub>m</sub>!

### 3.1 Discussion

The model presented here subsumes what has sometimes been referred to in the literature as cases of 'anaphoric destressing' (Liberman 1978), 'deaccenting' (Bolinger 1972, Chafe 1974) or 'default accent' (Ladd 1978). The term 'deaccenting' seems conceptually inappropriate, since it implies a negative process, e.g. lowering pitch, whereas the approach taken here is to view sentence stress as the **addition** of some parameter(s) of phonetic prominence to an already existing word stress.<sup>6</sup> Deaccenting is assumed to apply in the following examples from Ladd (1978:52–3):

- (23) A: John was mad because he got nothing but books for Christmas.  
B: Oh, doesn't John **read** books?  
(24) Harry wants a VW but his wife would prefer an **American** car.

where *books* in (23B) and *car* in (24) are assumed to be deaccented by some means or another. 'Default accent' is a better term, but what it describes is the natural output of the model presented here, i.e. sentence stress in a dialogue context. Note that according to our model, the sentence stress on

<sup>6</sup> This is the position taken, for example, by Bruce (1977).

The hierarchy in (16) can account for the troublesome examples for Bresnan's and Berman and Szamosi's accounts of sentence stress such as

where *plans* is the object (stress according to step 1 in Figure 1),

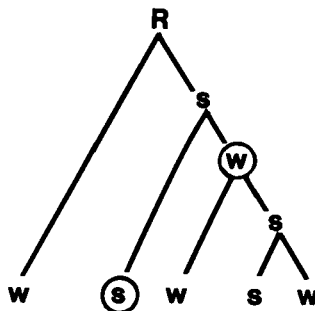
where *to leave* is the object, *has plans* being analyzed as a complex verb with the meaning ‘intends’ (cf. Chomsky and Halle 1968:24) (stress according to step 1 in Figure 1), and

where *clams* is the object (stress according to step 1 in Figure 1). The model presented here is more general than previous models of sentence stress in that it can also explain examples accounted for by the NSR, Bresnan's Topicalization Rule, and Berman and Szamosi's Surface Structure Generalization, e.g.:

(10) Mary liked the **proposal** that was made (SS Generalization)

(13) (f) **Guests** arrived (Topicalization Rule)

Notice that the present model also explains certain problematic cases for Liberman's (1978) metrical analysis of sentence prosody, e.g. (25)



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With respect to this example, Liberman (1978:167–8) makes the following remarks:

‘Presumably the reason that the constituent which contains the circled nodes has the form [sw], instead of the “null hypothesis” form [ws], is that anaphoric or redundant material is considered intrinsically **weak**, as opposed to anything that adds new information to the conversation. The other side of the coin is obviously that crucial or extra-important new information will want to be **strong**. Often it is hard to determine, in a particular case, what represents the result of destressing old information and what represents the result of stressing new information. One not unreasonable hypothesis would permit free assignment of either [ws] or [sw] to any phrasal constituent, but would stipulate that the pattern [sw] will be assumed to have some special pragmatic justification. The alternative is to postulate some set of features like [ $\pm$ contrastive], assigned to nodes, which influence prosodic and tonal phenomena in some way. The main difficulty in this area is that nobody has ever gotten the facts straight, to my knowledge.’

The fact that the verb *hit* receives sentence stress in Liberman’s example is a natural outcome of the model presented here: both the object and subject are contextually coreferent<sup>5</sup> and consequently stress is assigned according to step (3a) in Figure 1.

The biggest problem in the application of our model lies in determining coreferential relations. In cases where anaphoric pronouns are used, the situation is very clear, e.g.:

- (26) (a) Rats<sub>i</sub> like **cheese**<sub>j</sub>  
(b) It’s<sub>j</sub> **good** for them<sub>i</sub>

However, in more grammatically opaque cases of coreference or cases involving lexical relations such as synonymy, hyponymy and part-whole relations, the situation becomes more complicated, both for the listener in a concrete speech situation and for the linguist in attempting to account for the conditions under which such contextual identity is made. Consider, for example, the following two sentences uttered by Kurt Vonnegut in an interview on Swedish Radio:

- (27) (a) When you laugh or cry, you are in fact throwing off (all sorts of things which have been released into (your bloodstream)<sub>i</sub>); which will allow you to run or to stand and fight.  
(b) And if you can do neither, then (your body)<sub>i</sub> has to get **rid** of (these chemicals)<sub>j</sub>:

In this case, *these chemicals* in (27b) is coreferent with the phrase *all sorts of things which have been released into your bloodstream* and *your body* is related in a part/whole relation to *your bloodstream* in (25a). Note that it is the result of the speaker's placing sentence stress on the verb, *get rid* that we, as listeners, are able to infer such identity relations between what is being said and what has previously been said.<sup>7</sup>

Whether the concept of contextual coreference can be further developed in order to account for other cases of sense relationships involving coreference beyond the lexical level is an open question. For instance, in order to account for the so-called 'contrastive stress' patterns in sentences such as (28) (see Ladd 1978:78 ff.), the relation of coreference must be extended to cover identity of reference of semantic roles.

(28) Peter<sub>i</sub> kicked Ulrika<sub>j</sub> and then **she**<sub>j</sub> kicked **him**<sub>i</sub>

In (28), the conditioning environment for sentence stress lies, not on the level of lexical referential identity, but rather on the level of identity of semantic roles, the stress pattern in this case serving to indicate a change in the 'actor/patient' relationship (see Enqvist 1979:140–1). It is possible that such nonlexical coreferential relations could be built into a surface syntactic model to the extent that they can be related to a change of grammatical functions in otherwise referentially identical surface structure strings. As it stands now, however, our model cannot account for such cases.

In conclusion, we should note that the dynamics of sentence stress is an issue of current interest in research on speech production and comprehension. We feel that the analysis presented here provides an insight into the direction in which further work in this area can proceed. Our model is, for example, well suited for application in computer models of verbal production such as that presented in Sigurd (1982).

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<sup>7</sup> See Nooteboom and Terken (1982) for experimental support for this notion.

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